Composition and the Algebra of Mechanisms

Mechanisms as stateful transducers over distributional beliefs

Peter Cotton · *Working draft v0.2* · 2026

Abstract

Scoring rules, market makers, parimutuels, and opinion pools are usually presented as separate mechanisms. This note studies how they compose when each stage is treated as a stateful transducer whose common message type is a distributional belief. The incentive-bearing core is generated by convex potentials: proper scores arise from convex entropies, Hanson's sequential scoring-rule construction gives cost-function market makers by Fenchel conjugacy, and CFMMs are related to cost-function markets by convex level-set dualities. A small operator set (sequentialise, pool, conjugate, ensemble, residual) organizes the catalogue, and the pooling and merging operators are theorems: the linear and logarithmic opinion pools are the two Kullback-Leibler barycenters, and merging two market makers is an infimal convolution under which liquidity adds. A worked pipeline chains an elicitation market into a PIT-based calibration critic, with wealth updated by proper-score increments. The note separates four issues that are often blurred: closure of the message type, convex generation of the individual mechanisms, propriety of transformed scores, and incentive compatibility of multi-stage strategic composition. The first three admit theorems, stated here with proofs; the fourth is open beyond special cases.

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1. Introduction

Mechanisms for eliciting and aggregating forecasts are usually studied one at a time. A proper scoring rule is analysed as a one-shot contract; a market maker as a sequential trading venue; an opinion pool as an estimator; a calibration test as a diagnostic. Real systems chain them. A forecasting platform elicits distributional predictions, aggregates them into a consensus, tests the consensus for calibration, and routes rewards back to the participants, all in a loop. The microprediction platform ran exactly this loop at scale (Cotton 2022), and the stacked-lottery design behind it was presented at MIT CSAIL in 2020 (Cotton 2020, slides 29-31): market-implied percentiles are themselves the subject of further games, and calibration is achieved by composing monotone transformations, each contributed by one or more algorithms.

Composition is also where guarantees break. Each stage of such a pipeline can be individually truthful while the chain is not. The companion paper on point-cloud scoring gives a concrete example: inserting an innocuous-looking kernel-density smoothing step between the submission and the score changes the optimal report from the forecaster's belief to its deconvolution. Seams matter, so it is worth being precise about what survives them.

This note organizes the catalogue around four questions that are often blurred together:

1. *Message closure.* Can every stage consume and emit one common object, so that stages plug together at all?

2. *Convex generation.* Is each individual mechanism generated by a convex potential, so that one dictionary covers scoring rules, market makers, and their duals?
3. *Propriety under transformation.* When a stage transforms forecasts or outcomes, does strict propriety survive?
4. *Strategic composition.* When stages couple participants' payoffs, do the single-stage incentive properties survive equilibrium play?

The first is an engineering convention (§2, §6). The second and third admit theorems, which are stated with complete proofs in §3 and §4; none of the mathematics is new, but the proofs are collected in one place and one convention. The fourth is open beyond the special cases analysed in the companion paper, and §7 records what is known.

2. Preliminaries

The finite setting. Sections 3 and 4 work with a finite outcome set $\{1, \dots, n\}$ and reports p in the probability simplex Δ. This is the cleanest setting for the convex dictionary. The implementation-level object in the repository is continuous, a weighted Gaussian mixture exposing `.mean`, `.std`, `.quantile`, `.logpdf`, `.cdf`; the corresponding continuous theory uses proper scoring rules on probability measures (log score, CRPS, energy and kernel scores) or discretized approximations, and the finite statements below transfer with the usual measure-theoretic care. We flag the one place where the transfer is not routine (the PIT, Proposition 4).

Scores. A scoring rule assigns $S(p, i) \in \mathbb{R} \cup \{-\infty\}$ to a report p and outcome i; the expected score of reporting p under belief q is $S(p; q) = \sum_i q_i S(p, i)$, assumed well defined with $S(q; q)$ finite (this is the regularity used throughout). The rule is *proper* if $S(q; q) \geq S(p; q)$ for all p, q and *strictly proper* if equality forces $p = q$. Conventions are reward-form (higher is better); the repository's implementations use the loss form.

Convex tools. For convex G with subgradient selection $G'(p) \in \partial G(p)$, the Bregman divergence is $D_G(q, p) = G(q) - G(p) - \langle G'(p), q - p\rangle$. The Legendre-Fenchel conjugate of a function R on Δ is $R^*(\mathbf{q}) = \sup_{p \in \Delta} \langle p, \mathbf{q}\rangle - R(p)$. The infimal convolution of f and g is $(f \,\square\, g)(x) = \inf_y f(y) + g(x - y)$. "Closed proper convex" is used in the standard sense (Rockafellar 1970). Differentiable statements are on the relative interior of Δ, with extended-real values allowed on the boundary.

3. One potential, three mechanisms

Theorem 1 (characterisation of proper scoring rules; Savage (1971), McCarthy (1956)). *A regular scoring rule S is proper iff there is a convex $G : \Delta \to \mathbb{R}$ with*

$$S(p, i) \;=\; G(p) + \langle\, G'(p),\; e_i - p\,\rangle, \qquad G'(p) \in \partial G(p),$$

where e_i is the i-th unit vector. It is strictly proper iff G is strictly convex relative to Δ, and then $G(p) = S(p; p)$ is the expected score of a truthful forecaster.

Proof. ($\Leftarrow$) Since $\sum_i q_i(e_i - p) = q - p$, the expected score is affine in the belief: $S(p; q) = G(p) + \langle G'(p), q - p\rangle$. Hence

$$S(q;q) - S(p;q) \;=\; G(q) - G(p) - \langle G'(p),\, q-p\rangle \;=\; D_G(q,p),$$

which is non-negative by the supporting-hyperplane inequality, and zero only at $q = p$ when G is strictly convex relative to Δ; so S is (strictly) proper.

($\Rightarrow$) Define $G(q) := S(q;q)$. Properness says $G(q) = \sup_p S(p;q)$, and each $q \mapsto S(p;q) = \langle S(p,\cdot), q\rangle$ is affine, so G is a pointwise supremum of affine functions, hence convex. For any fixed p,

$$S(p;q) \;=\; G(p) + \langle S(p,\cdot),\, q-p\rangle \quad \text{with} \quad S(p;q) \le G(q) \;\; \forall q,$$

so the vector $S(p,\cdot)$ is a subgradient of G at p (acting on the tangent space of Δ, where subgradients are defined up to a constant shift along $\mathbf{1}$, which the representation absorbs). Substituting $G'(p) = S(p,\cdot)$ into the display returns $S(p,i)$ identically. If S is strictly proper the supremum has a unique maximiser at every q, which for a supremum of affine functions is equivalent to strict convexity relative to Δ. ■

The content of Theorem 1 is the dictionary *proper scoring rule $\leftrightarrow$ convex function $\leftrightarrow$ Bregman divergence* (Gneiting and Raftery 2007; Banerjee et al. 2005). The classics are three choices of G, with the divergences computed directly from the definition:

Score	Generator $G(p) = S(p;p)$	Bregman divergence $D_G(q,p)$
logarithmic	$\sum_i p_i \log p_i$	$\mathrm{KL}(q\\|p)$
Brier (quadratic)	$\\|p\\|_2^2$	$\\|q-p\\|_2^2$
spherical	$\\|p\\|_2$	$\\|q\\|_2\,(1-\cos\theta_{p,q})$

For the spherical row: $\nabla G(p) = p/\|p\|$, so $D_G(q,p) = \|q\| - \langle p,q\rangle/\|p\| = \|q\|(1-\cos\theta_{p,q})$, an angular term scaled by $\|q\|$. For the logarithmic row the representation gives $S(p,i) = \log p_i$ on the relative interior, with $-\infty$ on the boundary.

Theorem 2 (scoring rule to market maker; Hanson (2007), Abernethy et al. (2013)). *Let R be closed, proper, and strictly convex on Δ (in the sequel, $R = G$, the generator of Theorem 1 read as a regulariser), and define*

$$C(\mathbf{q}) = \sup_{p\in\Delta} \big(\langle p, \mathbf{q}\rangle - R(p)\big).$$

Then:

(i) C is convex and finite, the maximiser $p^*(\mathbf{q})$ is unique, and $\nabla C(\mathbf{q}) = p^*(\mathbf{q}) \in \Delta$: prices are a probability vector. (Without strict convexity, prices live in $\partial C(\mathbf{q})$.)

(ii) C is translation-equivariant, $C(\mathbf{q}+\alpha\mathbf{1}) = C(\mathbf{q}) + \alpha$; costs telescope over any trade path, so round trips cost zero and buying the full bundle costs its payout: the market is arbitrage-free.

(iii) With initial state $\mathbf{q}_0 = \mathbf{0}$, the maker's loss when the market settles on outcome i after net sales $\mathbf{q}$ is

$$\mathrm{loss}_i(\mathbf{q}) = q_i - C(\mathbf{q}) + C(\mathbf{0}) \;\le\; R(e_i) - \inf_{p\in\Delta} R(p) \;\le\; \sup_{p\in\Delta} R(p) - \inf_{p\in\Delta} R(p).$$

(iv) Taking $R(p) = b\sum_i p_i \log p_i$ gives $C(\mathbf{q}) = b\log\sum_i e^{q_i/b}$, Hanson's LMSR, with worst-case loss $b\log n$.

Proof. (i) C is a supremum of affine functions of $\mathbf{q}$, hence convex; the supremum of a continuous function over the compact Δ is attained, and strict concavity of $p \mapsto \langle p, \mathbf{q}\rangle - R(p)$ makes the maximiser unique. Danskin's theorem gives $\nabla C(\mathbf{q}) = p^*(\mathbf{q})$. (ii) $\langle p, \mathbf{q} + \alpha\mathbf{1}\rangle = \langle p, \mathbf{q}\rangle + \alpha$ for $p \in \Delta$, so the supremum shifts by α. A trader moving the state $\mathbf{q} \to \mathbf{q}'$ pays $C(\mathbf{q}') - C(\mathbf{q})$ by definition, so costs over any path telescope; a round trip costs zero, and by translation equivariance the bundle $\alpha\mathbf{1}$ costs exactly α, its sure payout. (iii) The maker collects $C(\mathbf{q}) - C(\mathbf{0})$ and pays q_i, so $\mathrm{loss}_i = q_i - C(\mathbf{q}) + C(\mathbf{0})$. By Fenchel-Moreau, $\sup_{\mathbf{q}}\left(\langle e_i, \mathbf{q}\rangle - C(\mathbf{q})\right) = R^{**}(e_i) = R(e_i)$, and $C(\mathbf{0}) = \sup_p -R(p) = -\inf_p R(p)$. The final inequality holds because $e_i \in \Delta$; for convex R the supremum over Δ is attained at an extreme point, so it is typically an equality. (iv) Lagrange: maximising $\langle p, \mathbf{q}\rangle - b\sum p_i \log p_i$ subject to $\sum p_i = 1$ gives $q_i - b(\log p_i + 1) = \lambda$, so $p_i \propto e^{q_i/b}$, and substituting back yields $C(\mathbf{q}) = b\log\sum_i e^{q_i/b}$. Then $R(e_i) = 0$ and $\inf R = -b\log n$ at the uniform distribution, so the loss bound is $b\log n$. ■

Proposition 3 (cost-function markets and CFMMs). *Under the monotonicity, concavity, and reserve-domain hypotheses of Frongillo et al. (2024), cost-function prediction markets and constant-function market makers can be converted into one another. In the unconstrained-reserve sign convention a cost function C gives a concave CFMM potential by $\varphi(\mathbf{r}) = -C(-\mathbf{r})$; bounded-reserve versions require a perspective (level-set) construction. The equivalence is a convex level-set duality; it is not the bare Fenchel conjugacy $C \mapsto C^*$.*

We do not reprove the general equivalence here; Frongillo et al. (2024) give it in full, and Angeris et al. (2023) and Angeris et al. (2021) develop the CFMM side. What the note uses is the direction that convex analysis computes cleanly, illustrated on the canonical example.

Example (constant product). Take two outcome tokens with reserves r_1, r_2 and invariant $r_1 r_2 = k$. The pool's portfolio value at prices $(p, 1-p)$ is

$$V(p) = \inf\{p\, r_1 + (1-p)\, r_2 :\; r_1 r_2 = k\} = 2\sqrt{k\, p(1-p)},$$

by the AM-GM inequality, with the infimum attained at $r_1 = \sqrt{k(1-p)/p}$. V is concave; reading $R(p) = -V(p)$ as the regulariser of Theorem 2 gives a maker whose worst-case loss is the range of R on $[0,1]$, namely $\sqrt{k}$ (between $p = \frac{1}{2}$ and the endpoints): the geometric mean of the initial reserves. The corresponding generator is not entropic, which is the convex-analytic content of the observation that constant-product markets and the LMSR price bounded-payout claims differently.

Proposition 4 (the probability integral transform; Rosenblatt (1952), Dawid (1984)). *If X has continuous CDF F then $U = F(X) \sim \mathrm{Uniform}(0,1)$, and $z = \Phi^{-1}(U) \sim N(0,1)$. In the prequential setting, if F_t is the true conditional law of X_t given the past, then $U_t = F_t(X_t)$ is conditionally uniform and the PIT stream is iid uniform.*

Proof. For continuous F, with the generalized inverse $F^{-}(u) = \inf\{x : F(x) \geq u\}$, the event $\{F(X) \leq u\}$ differs from $\{X \leq F^{-}(u)\}$ by a set of probability zero, and $\Pr(X \leq F^{-}(u)) = F(F^{-}(u)) = u$ by continuity. The prequential statement applies this conditionally at each t. ■

Two scope remarks, both load-bearing. First, the converse fails in the strong sense: marginally uniform PITs do not certify an informative forecast. Reporting the unconditional law F of an iid sequence gives $F(X_t) \sim \mathrm{Uniform}(0, 1)$ even if valuable covariates were ignored. A PIT critic therefore witnesses miscalibration relative to its test class; it complements, and does not replace, a proper score that rewards sharp conditional distributions. Second, for discrete forecasts the randomized PIT preserves the exact uniform null, while the mid-PIT is a convenient deterministic diagnostic with a different null distribution; the demo of §5 uses the mid-PIT and should be read accordingly.

4. Operators on mechanisms

The common signature. The mechanisms can be organized as a typed algebra of stateful transducers. The common message type is a distributional belief; wealth and transfers are side channels. A stage is a map

$$M : \ (\text{Dist}^m, \ w, \ x) \ \longmapsto \ (\text{Dist}, \ w', \ \pi),$$

taking m participant reports, a wealth state w, and (when available) a realized outcome x, and returning an aggregate belief, an updated wealth state, and transfers $\pi \in \mathbb{R}^m$. A scoring rule is the transfer map inside such a stage, not itself a Dist-to-Dist morphism; a market maker is a stage whose internal state is the inventory vector and whose emitted Dist is the price; an opinion pool is a stage with no outcome argument and zero transfers. Composition means wiring the Dist output of one stage to the Dist inputs of the next while the side channels thread through.

The operators below act on stages. The first two are constructions from a score; the next two are theorems about aggregation and merging; the fifth is where propriety can fail; the last two are directions.

Sequentialise. Theorem 2: a proper score run sequentially against a wealth state is a cost-function market maker.

Pool. A proper score gives a batch elicitation mechanism when reports are scored independently and funded externally: properness is inherited report by report. Parimutuel and budget-balanced versions are a different game, because the pot split couples payoffs through the denominator; the companion paper's §2 gives the price-taking analysis (truthful all-in, a symmetric equilibrium at fractional stakes, degenerate as the stake fraction vanishes), and beyond that the equilibrium theory is open.

Ensemble (Proposition 5: the two pools are the two KL barycenters). *Let $q_1, \ldots, q_m$ be densities and $w_i \geq 0$, $\sum w_i = 1$. Then the linear pool minimizes the forward divergences and the logarithmic pool the reverse:*

$$\arg\min_p \textstyle\sum_i w_i \, \mathrm{KL}(q_i \| p) = \sum_i w_i q_i, \qquad \arg\min_p \sum_i w_i \, \mathrm{KL}(p \| q_i) \ \propto \ \prod_i q_i^{w_i}.$$

Proof. For the first, $\sum_i w_i \, \mathrm{KL}(q_i \| p) = \text{const} - \int (\sum_i w_i q_i) \log p$, and $\int \bar{q} \log p$ is maximized over densities at $p = \bar{q}$ (Gibbs). For the second, $\sum_i w_i \, \mathrm{KL}(p \| q_i) = \int p \log p - \int p \, \overline{\log q}$ with $\overline{\log q} = \sum_i w_i \log q_i$; the Lagrange condition is $\log p = \overline{\log q} + \text{const}$. ■

One distinction matters in practice. Training weights by cumulative log score and then mixing linearly, $\sum_m w_m F_m$, is Bayesian model averaging, a linear pool with score-trained weights; the logarithmic pool multiplies densities and renormalizes. Both appear in the repository's `aggregation` module; they are different aggregates with different sharpness (Genest and Zidek 1986).

Merge (Proposition 6: merging makers is infimal convolution). *For closed proper convex f, g: $(f \,\square\, g)^* = f^* + g^*$. Consequently, merging two cost-function makers with regularisers R_1, R_2 (cost functions $C_i = R_i^*$) yields the maker with regulariser $R_1 + R_2$, and merging* LMSR_{b_1} *with* LMSR_{b_2} *yields* $\mathrm{LMSR}_{b_1+b_2}$*: liquidity adds.*

Proof. $(f \,\square\, g)^*(p) = \sup_x \langle p, x \rangle - \inf_y \{ f(y) + g(x-y) \} = \sup_{y,z} \langle p, y \rangle - f(y) + \langle p, z \rangle - g(z) = f^*(p) + g^*(p)$. For the makers, $C_1 \,\square\, C_2 = (R_1 + R_2)^*$ by biconjugacy, and $b_1 G + b_2 G = (b_1 + b_2) G$ for the entropic generator. ■

This is also the risk-sharing literature's composition law: the aggregate of several agents' convex risk measures is their infimal convolution, with the Pareto allocation as minimiser and the common subgradient as the clearing price (Barrieu and El Karoui 2005; Jouini et al. 2008); the market-liquidity reading is developed by Bhaskara et al. (2023).

Conjugate. Run a stage in a transformed coordinate and map back. This is where propriety needs care, and the theory splits in two:

- *Fixed transformations.* Properness is preserved under any fixed transformation of forecast and outcome, and strict propriety iff the transformation is injective (Allen et al. 2023, Prop. 4; Pic et al. 2025, Prop. 1). The Markov-kernel (stochastic channel) extension, strict propriety iff the channel is injective on laws, is Theorem 2 of the companion point-cloud paper, whose Theorem 1 also exhibits the canonical failure: a KDE smoothing seam whose raw-outcome score elicits a deconvolution.
- *Forecast-dependent transformations.* The PIT critic transforms outcomes by the reported F itself, so the fixed-transformation theorems do not apply to it; Proposition 4 and its scope remarks are the correct warrant, and the critic is a calibration diagnostic rather than a strictly proper score. The stacked-lottery design (Cotton 2020, slides 29-31) is this operator in practice: percentiles from one game feed the next, and calibration is produced by composing monotone maps contributed by competing algorithms.

Residual. One stage models what the previous stage got wrong: a correction (boosting) market on the residual stream. Named here as an operator; the construction and its incentive analysis are open (§7).

Spec. Serialise a pipeline to data and search over it; the mechanism analogue is a market over pipelines. Also open.

5. Worked composition: an elicitation market feeding a calibration critic

The simplest non-trivial pipeline chains two stages through a feedback loop (runnable: `examples/sim_pipeline.py`):

1. *Elicitation market.* A pool of forecasters each report a predictive distribution; the wealth-weighted linear opinion pool is a single predictive distribution F_t, the transformation that should turn outcomes into noise.
2. *Calibration critic.* Apply the PIT, $u_t = F_t(x_t)$; if F_t is the true conditional law the u_t are iid uniform (Proposition 4). The critic measures the departure of the PIT stream from uniformity; its detectable edge is a witness to miscalibration relative to its test class.
3. *Feedback.* Each forecaster's wealth is updated by its log score. Wealth is the score-driven credit state: score increments provide the update signal, wealth the multiplicative credit variable, and the aggregate F_t sharpens as credit flows to calibrated reports.

This is a market-native GAN in loose analogy: a generator (the transform market) against a critic (the uniformity market), with a proper score as the loss. The analogy is organisational, not an equivalence.

The demo makes the loop quantitative. Five forecasters (two calibrated, plus a bull, a bear, and an overconfident report) against an iid $N(0,1)$ stream:

- PIT-stream uniformity error (total-variation distance from uniform): equal-weighted aggregate 0.165, converged aggregate 0.068;
- bias (mean z_t with $z_t = \Phi^{-1}(u_t)$) stays near zero throughout: the symmetric bull and bear cancel, so the mean never reveals the problem. It is the shape of the PIT stream that is wrong and then fixed, which is why the critic tests the whole stream rather than a moment or two;
- wealth on the two calibrated reports climbs $0.41 \to 1.00$ as the loop runs.

The demo uses the mid-PIT for the discrete grid, so its uniformity numbers are diagnostic approximations in the sense of §3.

6. Implementation

The operational discipline is borrowed from the skaters time-series interface (Cotton 2026), a dependency-free successor to timemachines (Cotton 2021). A skater has the contract

```
dists, state = f(y, state)          # dists : list[Dist] for horizons 1..k
```

and every payload is the same `Dist`. Transforms, ensembles, and search all consume and emit that type; that closure is the whole reason a handful of operators suffice. The mechanism analogue threads wealth where a skater threads model state:

```
payouts, aggregate, wealth = M(reports, wealth, outcome)
#   reports   : list[Dist]   participants' beliefs      (skater's y)
#   wealth    : state        incentive-bearing          (skater's state)
#   aggregate : Dist         price / implied prob       (skater's dists)
```

which is the signature of §4 in code. A market maker is a skater whose state is the inventory vector and whose emitted `Dist` is the price; Hanson's construction (Theorem 2) is the skater move applied to a score.

7. Open problems

1. *Strategic composition.* Individually truthful stages can admit strategic play across the chain. The companion paper settles the price-taking pot split (truthful all-in, symmetric equilibrium at fractional stakes, degenerate at vanishing stakes); the finite-player, multi-stage equilibrium theory is open.
2. *Conservation.* Wealth is the threaded state and budget balance is free stage by stage; the substantive invariant is whether the elicited edge (the Bregman divergence a truthful participant holds) telescopes across stage seams. The companion research note on conservation and boosting states the conjecture: edge is conserved iff the interface is a sufficient (injective) reduction.
3. *Residual markets.* A chain of markets in which stage t prices the residual of stage $t-1$, with wealth as credit routing, remains to be constructed and analysed against the boosting literature.

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Runnable demo: `examples/sim_pipeline.py`. Companion papers and notes: Scoring Point-Cloud Distributional Submissions; conservation and boosting; the relationship map.